Fiscal Year:	FY 2011	**Task Last Updated:**	FY 05/04/2011
PI Name:	Shoukas, Artin A. Ph.D.		
Project Title:	Radiation, Endothelial Cell Senescence, Accelerated Aging, and Atherosclerosis		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21205-2109	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 NSBRI-RFA-07-01 Human Health in Space
Start Date:	03/01/2008	**End Date:**	02/28/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Dr. Dan Berkowitz took over as PI sometime in 2012; see Berkowitz for subsequent reporting (Ed., 2/12/2013)		
COI Name (Institution):	Berkowitz, Dan (The Johns Hopkins University) Nyhan, Daniel (The Johns Hopkins University) Romer, Lewis (Johns Hopkins Medical School) DeWeese, Theodore (Johns Hopkins Medical School)		
Grant/Contract No.:	NCC 9-58-CA01301		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Exposure to ionizing radiation represents one of the most significant health risks to astronauts during both short, but particularly long, duration spaceflight. Vascular endothelial cells (ECs), the most radiosensitive vascular cell type, remain the major regulating interface between the blood and vessel wall. There, they act as a paracrine organ releasing signaling molecules that regulate vessels tone, permeability and cell and platelet adhesion. Dysfunction of the endothelium may result from both an increase in endothelial cell damage (oxidative stress with nitroso-redox imbalance), an impairment of the endothelial repair mechanisms (decrease in the number and function of endothelial progenitor cells) or an acceleration of endothelial cell senescence. Radiation is known to induce cellular oxidative stress, as well as suppress bone marrow proliferation. Data from our lab suggests that gamma and heavy iron ion radiation results in the activation of the ROS producing enzyme xanthine oxidase (XO). Activation of enzymes such as arginase results in a decrease in endothelial protective nitric oxide (NO) with altered nitroso-redox balance and endothelial damage. We hypothesize that radiation will result in activation of ROS producing enzymes that will result in increased oxidative stress, nitroso-redox imbalance and endothelial cell damage. We also hypothesize that radiation will decrease endothelial repair mechanism by reducing bone marrow derived endothelial progenitor cells (EPCs). Additionally, we hypothesize that radiation will accelerate endothelial cell senescence by decreasing telomerase activity and increasing the rate of telomere attrition. This will ultimately result in significant endothelial dysfunction, mimic the aging phenotype and accelerate the development of atherosclerosis. We will test the countermeasure hypothesis that the XO inhibitor (oxypurinol) will reduce radiation related endothelial dysfunction, that arginase inhibition (which reciprocally regulates nitric oxide synthase) will restore nitroso-redox balance and promote endothelial cell protection. We will also test the hypothesis that statins (endothelial cell protectants through a number of pleotropic effects independent of their lipid lowering effects) will prevent oxidative stress, improve nitroso-redox imbalance, enhance endothelial progenitor cell production, and prevent telomere decay. In this way, the adverse effects of radiation on the endothelial cell will be attenuated. In previous years we have demonstrated that radiation-induced endothelial dysfunction is largely due to the activation of xanthine oxidase in rat aorta as reactive oxygen species derived from xanthine oxidase contribute to impaired endothelial nitric oxide production and attenuated endothelial-dependent vasorelaxation. In the future, we hope to investigate further mechanisms contributing to vascular radiation injury through the use of knockout mouse models. Therefore, during the past year we have begun to characterize the vascular response to radiation of mice. In addition, we are interested in intrinsic endothelial defense mechanisms, and whether these mechanisms are dysregulated by radiation. Currently, are investigating the role of endothelial progenitor cells (EPCs) in vascular repair after low-dose ionizing radiation (IR) injury. Previous studies have suggested that locally applied low-dose IR is angiogenic. Epidemiological studies and studies from our lab, however, demonstrate that whole body IR exposure is anti-angiogenic and promotes endothelial dysfunction. We hypothesize that the discrepancy between the effects of local and total body irradiation is due to the degree of intact EPC function, where with local exposure, function is intact, compared with total body exposure, where EPC function is most likely compromised. We are investigating the effects of low-dose heavy ion radiation on EPC function and vascular repair using probabilistic and animal models.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The current work being performed has the potential to protect humans undergoing either diagnostic radiation procedures or radiation treatment for cancerous tumors. Development of countermeasures for the protection of astronauts may be applicable to prevent morbidities seen in patients undergoing radiation therapy for cancerous tumors.
Task Progress:	Currently, are investigating the role of endothelial progenitor cells (EPCs) in vascular repair after low-dose ionizing radiation (IR) injury. Previous studies have suggested that locally applied low-dose IR is angiogenic. Epidemiological studies and studies from our lab, however, demonstrate that whole body IR exposure is anti-angiogenic and promotes endothelial dysfunction. We hypothesize that the discrepancy between the effects of local and total body irradiation is due to the degree of intact EPC function, where with local exposure, function is intact, compared with total body exposure, where EPC function is most likely compromised. We are investigating the effects of low-dose heavy ion radiation on EPC function and vascular repair using probabilistic and animal models. We are still in the investigative stage to determine mechanisms of radiation damage and methodology to protect astronauts. NOTE: Dr. Dan Berkowitz took over as PI sometime in 2012; see Berkowitz for subsequent reporting (Ed., 2/12/2013)
Bibliography Type:	Description: (Last Updated: 05/05/2011)
Articles in Peer-reviewed Journals	Soucy KG, Attarzadeh DO, Ramachandran R, Soucy PA, Romer LH, Shoukas AA, Berkowitz DE. "Single exposure to radiation produces early anti-angiogenic effects in mouse aorta." Radiat Environ Biophys. 2010 Aug;49(3):397-404. Epub 2010 Apr 18. PMID: 20401726 , Aug-2010
Articles in Peer-reviewed Journals	Soucy KG, Lim HK, Attarzadeh DO, Santhanam L, Kim JH, Bhunia AK, Sevinc B, Ryoo S, Vazquez ME, Nyhan D, Shoukas AA, Berkowitz DE. "Dietary inhibition of xanthine oxidase attenuates radiation-induced endothelial dysfunction in rat aorta." J Appl Physiol. 2010 May;108(5):1250-8. Epub 2010 Feb 18. PMID: 20167676 , May-2010